Ismail Chekalil

Mechanical Eng. PMP. PhD. R&D.

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KEY SKILLS

- Metal & Composite Additive Manufacturing
- Welding (Friction stir welding)
- Sustainable Manufacturing
- Mechanical Characterization
- Material science
- Manufacturing Process Monitoring
- Teaching and mentoring
- Project management
- CNC Machining

SUMMARY

Mechanical engineering researcher (Ph.D.) focused on manufacturing and materials, specializing in metal additive manufacturing and friction stir welding (FSW) with an emphasis on corrosion. Manage additive manufacturing and subtractive manufacturing labs, mentor students, and collaborate on sustainable manufacturing projects. Published several papers on FSW, corrosion behavior, and sustainability.

EXPERIENCE

Technical Research Assistant **Jan 2024 – Present**
Interdisciplinary Research Center - Intelligent Manufacturing and Robotics
King Fahd University of Petroleum and Minerals (67 worldwide ranking) , Saudi Arabia

- Pioneering Additive Manufacturing Processes: Utilize advanced metal, composite, and polymer 3D printing technologies, including Selective Laser Melting (SLM 280), Direct Energy Deposition (Meltio Robot Cell), and Fused Deposition Modeling (Creatbot D1000, D600, Makerbot, Prusa), to optimize production outcomes.
- Training and Mentorship: Guide and support students and researchers in developing advanced skills in additive manufacturing (metal, composite, and polymers) and Design for Manufacturing.
- Laboratory Management and Oversight: Administer the operational and research activities of the Additive Manufacturing and Subtractive Manufacturing Labs, ensuring alignment with institutional goals.
- Research Facilitation and Innovation in sustainable manufacturing: Collaborate on high-impact research projects focusing on sustainable manufacturing.
- Precision CNC Machining and monitoring: Oversee complex CNC operations on a 4-axis lathe (Index B400, Siemens 840sl control) by designing and implementing precise machining strategies using NX CAM software. Plus, monitoring the process using sensors and a camera.

Mechanical Engineer **Dec 2022 – Dec 2023**
Technological Platform – Industrial Technology, Algeria

- Project Management and Machine Installation: Supervised the installation and commissioning of diverse industrial machinery, equipment, and software, ensuring seamless transitions to operational readiness for projects exceeding $2 million.
- Advanced CNC Operations: Programmed, operated, and oversaw tooling procurement for a 5-axis milling machine (HERMLE C400, Heidenhain 640 control).
- Technical Training and Proficiency Development: Completed supplier-led training on cutting-edge technologies, including the HERMLE C400 milling center, KUKA laser welding robot, HEXAGON Absolute Arm, HEXAGON coordinate measuring machines, FRITSCH Laser Granulometry, PTV and Protomax waterjets, and SEI MERCURY laser cutting systems.
- CAD/CAM Design and Manufacturing: Applied CAD and CAM software to design components and generate precise machine codes for CNC machining, waterjet cutting, and sheet metal fabrication processes.
- Team Development and Knowledge Sharing: Delivered training sessions on machining, manufacturing techniques, and best practices to enhance team expertise.
- Engineering Consulting: Provided tailored engineering solutions and consultation services to industrial clients, addressing complex technical challenges and optimizing operations.

Part time Lecturer **2019 – 2022**
University of Chlef

- Course Delivery: Taught and facilitated coursework on Non-Destructive Testing (NDT), providing students with a comprehensive understanding of NDT techniques, applications, and industry standards. (PG)
- Skill Development: Guided students in mastering practical and theoretical aspects of NDT methods, including ultrasonic testing, radiography, magnetic particle testing, and dye penetrant inspection. (PG)
- Led practical sessions in physics, assisting students with hands-on experiments. (UG)

Part-Time Freelancer **2019 – 2023**
IKYTOOL

- CAD Design and 3D Printing Services: Delivered high-quality Computer-Aided Design (CAD) and 3D printing solutions to diverse clients, addressing unique design and manufacturing requirements.
- Reverse Engineering: Applied expertise in reverse engineering to recreate parts and mechanisms for prototype development, maintenance, and repair applications.
- Prototype Development: Collaborated with clients to design and fabricate prototypes, ensuring alignment with functional and aesthetic specifications.

SCIENTIFIC RESEARCH

Journal Articles

- "Effect of corrosive environments on the mechanical properties of the FSW joint of aluminum alloy AA3003". Ismail Chekalil et al. Journal of Materials Research and Technology, Volume 33, Pages 2353-2364 | Nov 2024 – Dec 2024https://doi.org/10.1016/j.jmrt.2024.09.167
- "Comparative study of ultrasonic and laser assisted machining for sustainable leather cutting in greener industry practices". S. Mekid, V. Swaminathan, I. Chekalil. Journal of Materials Research and Technology, Volume 33, Pages 2353-2364 | Nov 2024 – Dec 2024 https://doi.org/10.1038/s41598-025-05181-z
- "Hybrid deep learning model to predict the ultimate tensile strength of friction stir welded joints". Akshash M et al. Engineering Applications of Artificial Intelligence, Volume 154, 15 Aug 2025 https://doi.org/10.1016/j.engappai.2025.111001
- "Effect of Welding Parameters and Artificial Intelligence-Based Prediction of Maximum Temperature in Friction Stir Welding of AA3003 Alloy". A. Belaribi et al. The International Journal of Advanced Manufacturing Technology. 20 Oct 2025. https://doi.org/10.21203/rs.3.rs-7123190/v1
- "Predicting Damage in Steel Pipe Elbow Under Combined Pressure and Bending Using X-FEM". Chaaben A et al. International Journal of Steel Structures, 15 Apr 2025. https://doi.org/10.1007/s13296-025-00953-9
- "Corrosion behavior of AA3003 friction stir welded joints", I. Chekalil, R. Chadli, A. Ghazi, A. Miloudi, M.P. Planche, A. Amrouche. Colloids and Surfaces A: Physicochemical and Engineering Aspects, Volume 680, Jan 2024 https://doi.org/10.1016/j.colsurfa.2023.132673
- "Prediction of mechanical behavior of friction stir welded joints of AA3003 aluminum alloy", Ismail Chekalil, Abdelkader Miloudi, Marie-Pierre Planche, Abdelkader Ghazi. Frattura ed Integrità Strutturale, 2020, Volume 14(54), Pages 153–168. https://doi.org/10.3221/IGF-ESIS.

Conference Proceedings

- "Integration of coordinate measuring machines", Ismail CHEKALIL, Abdelkader MILOUDI, Mohamed MEGHIRA. National Congress on Mechanics, Materials and Metallurgy, Oran, Algeria, 28–29 May 2023.
- "Prediction of Microhardness Profile of Friction Stir Welded Joints of AA3003 Aluminum Alloy" Ismail Chekalil, Abdelkader Miloudi, Abdelkader Ghazi, Marie-Pierre Planche. International Conference on Technology, Engineering, and Science (IConTES 2022), Antalya, Turkey, 16–19 Nov 2023.
- "Études comparative gravimétrique et thermodynamique de la corrosion sur l'alliage d'aluminium 3003 soudé par friction malaxage dans un milieu acide" R. Chadli, I. Chekalil, A. Miloudi. 1er Séminaire International (Webinaire) Nouvelles Tendances Liées aux Matériaux de Construction Innovants, École Supérieure en Sciences Appliquées de Tlemcen, Algeria, 22–23 Mar 2022.
- "Effect of FSW welding parameters on mechanical behaviour and the quality of aluminium alloy 3003 joints" Ismail Chekalil, Abdelkader Miloudi, Abdelkader Ghazi. 2nd International Symposium on Materials Chemistry (ISyMC 2021), Université M'hamed Bougara Boumerdes, Algeria, 16–20 May 2021.
- "An investigation of friction stir welding parameters on microhardness across the various zones" Ismail Chekalil, Abdelkader Miloudi, Abdelkader Ghazi. Séminaire International sur l'Industrie et la Technologie, Oran, Algeria, Algerian Journal of Engineering, Architecture, and Urbanism (AJEAU), 12–13 Mar 2021.

Ongoing project

- Laser Wire DED: Dual-material additive manufacturing.
- Machining of metal additive-manufactured parts.
- Biochar for 3D printing.
- Lattice structure (selective laser melting).

EDUCATION

PhD in Mechanical Engineering - Manufacturing and Production
University of Djillali Liabes | 2023
Master's in Mechanical Engineering - Production Option
National Polytechnic School - Maurice Audin | 2018
Bachelor of Mechanical Engineering - Production Option
National Polytechnic School - Maurice Audin | 2018

CERTIFICATION

- Project Management Professional license (PMP) by the Project Management Institute (PMI). | 2025
- Metal Cutting Technology (SANDVIK) | 2024
- Training certification: Hexagon coordinate measuring machine| 2023
- Training certification: Hexagon absolute arm | 2023
- Training certification: SEI laser cut | 2023
- Training certification: ERM KUKA robot| 2023
- Training certification: Haidenhain 5axis center controle | 2022
- Scuba diving (P1 CMAS) controle | 2019

INTERNSHIP

- Piping Manufacturing Process Intern | SPA TubeMaghreb | 2017
- Sheet Metal Forming Intern | The National Company of Industrial Vehicles | 2016
- Polyester Production Intern | ENPC | 2015